

Authentication Monitoring & PowerShell Investigation

A hands-on Windows/Splunk investigation covering failed-logon correlation, alert creation, encoded PowerShell detection, script-block logging, and evidence-based analyst disposition.

2 SOC WORKFLOWS	5 FAILED LOGONS	4625 AUTH EVENT	1 + 4104 CORRELATED EVENTS
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ANALYST	Terrell Bilal	DATE	August 4, 2026
ENVIRONM ENT	Windows 11, Splunk Enterprise, Universal Forwarder, Sysmon	HOST	RELLSHELL17

Executive outcome

The lab produced two complete analyst workflows. First, five Windows failed-logon events were aggregated into a threshold-based Splunk alert. Second, an encoded PowerShell account-discovery command was reconstructed by correlating Sysmon process telemetry with PowerShell script-block logging.

FINAL DISPOSITION	Benign positive - authorized activity generated in a controlled home-lab environment. Detection logic and telemetry functioned as intended.
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Skills demonstrated

- SIEM search and SPL field extraction
 - Threshold detection and alert design
 - PowerShell 4104 content inspection
 - Time normalization troubleshooting
- Windows authentication analysis
 - Sysmon parent-child process review
 - MITRE ATT&CK technique mapping
 - Evidence-based incident disposition

01 Authentication Detection

Objective

Generate controlled failed logons, validate Windows Security Event ID 4625, aggregate repeated attempts in Splunk, and create a scheduled alert for five or more failures within a ten-minute window.

Field	Observed value	Analyst interpretation
Event ID	4625	An account failed to log on
Target user	SOC-Lab	Purpose-built local test account
Logon type	2	Interactive logon at the endpoint
Source IP	127.0.0.1	Local loopback; not a remote source
Status / substatus	0xC000006D / 0xC000006A	Authentication failure / incorrect password
Threshold	5 attempts	Detection condition met

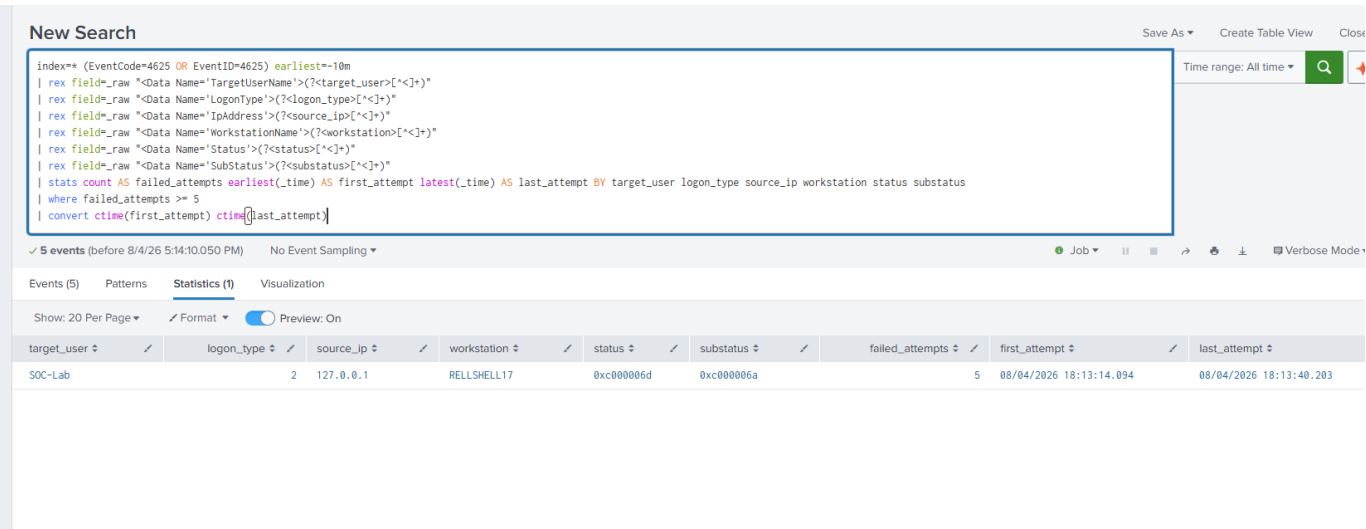


Figure 1. Five Event ID 4625 records correlated by account, source, workstation, status, and time window.

Detection logic

```
index=* (EventCode=4625 OR EventID=4625) earliest=-10m
| stats count AS failed_attempts earliest(_time) AS first_attempt latest(_time) AS last_attempt BY target_user logon_type source_ip
workstation status substatus
| where failed_attempts >= 5
```

The search was saved as a medium-severity scheduled alert named Multiple Failed Login Attempts. The rule runs every five minutes and triggers when at least one correlated result meets the threshold.

Troubleshooting note

An initial clock mismatch caused fresh events to appear outside the expected search window. Windows time synchronization corrected collection timing. This demonstrated why consistent endpoint and SIEM time settings are essential for reliable incident timelines.

02 PowerShell Threat Hunting

Objective

Detect an encoded PowerShell launch, recover the readable command through script-block logging, and correlate process telemetry with executed content in Splunk.

Controlled technique

A harmless command enumerated local accounts and their enabled state. It was Base64 encoded and executed with PowerShell's -EncodedCommand option to reproduce an obfuscation pattern commonly investigated by SOC analysts.

Get-LocalUser | Select-Object Name, Enabled

Source	Event	Evidence supplied
Sysmon Operational	1	powershell.exe process creation, parent process, user, integrity level, hashes, encoded command line
PowerShell Operational	4104	Decoded script block: Get-LocalUser Select-Object Name, Enabled

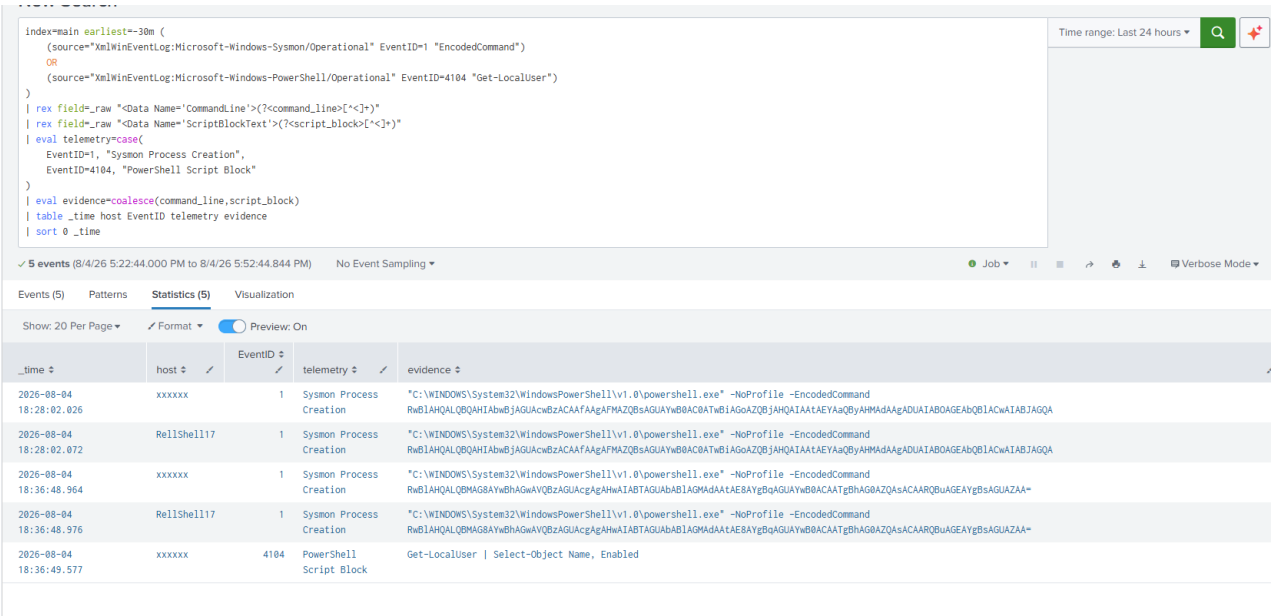


Figure 2. Correlated timeline: encoded Sysmon process telemetry followed by decoded Event ID 4104 content.

MITRE ATT&CK mapping

The activity maps to T1087.001 - Account Discovery: Local Account. The encoded PowerShell execution is a suspicious indicator, but it is not sufficient by itself to prove compromise. The decoded command, user context, source host, execution time, and authorization record determine disposition.

Configuration work

PowerShell Script Block Logging was enabled through the Windows policy registry and the PowerShell Operational event channel. Splunk Enterprise was then configured to ingest that channel, resolving the initial visibility gap where Event ID 4104 existed locally but was absent from the SIEM.

Findings, Disposition & Interview Story

Analyst findings

- Five incorrect passwords against SOC-Lab produced a threshold-based authentication finding.
- The failures were local interactive attempts from 127.0.0.1, not remote authentication attempts.
- Sysmon recorded the encoded PowerShell process and its parent-child execution context.
- Event ID 4104 revealed the decoded Get-LocalUser command, allowing the activity to be understood without manually trusting or executing an unknown payload.
- Duplicate Sysmon rows carried different host labels, indicating overlapping ingestion or host-label configuration rather than duplicate execution. This should be normalized before using event counts for production alert thresholds.

Disposition

CLASSIFICATION	Benign positive
RATIONALE	Activity was intentionally generated by the authorized analyst in a controlled home lab. Telemetry and detection logic operated as designed.
PRODUCTION ACTION	Document and close the authorized test. For an unrecognized occurrence, validate the user and host, decode or recover script content safely, review parent-child processes and network activity, and escalate if authorization cannot be confirmed.

Interview-ready explanation

I built a Splunk detection for repeated Windows authentication failures and then performed a PowerShell threat-hunting investigation. I enabled script-block logging, ingested Event ID 4104, and correlated it with Sysmon Event ID 1. Sysmon showed an encoded PowerShell launch while 4104 revealed the decoded Get-LocalUser command. I mapped the behavior to MITRE ATT&CK T1087.001, validated the user, host, source, and timing, and documented it as an authorized benign positive. During the lab I also resolved timestamp and ingestion gaps and identified duplicate event collection that could inflate alert counts.

Job-aligned capabilities

SIEM monitoring	Alert triage	Windows log analysis	Detection engineering
Threat hunting	Incident documentation	MITRE ATT&CK	Endpoint telemetry